

طراحی الگوریتم ها

ملکی مجد

مباحث

مساله بیشترین جریان (شار)

Maximum Flow

Max-Flow Min-Cut Theorem

Ford-Fulkerson method

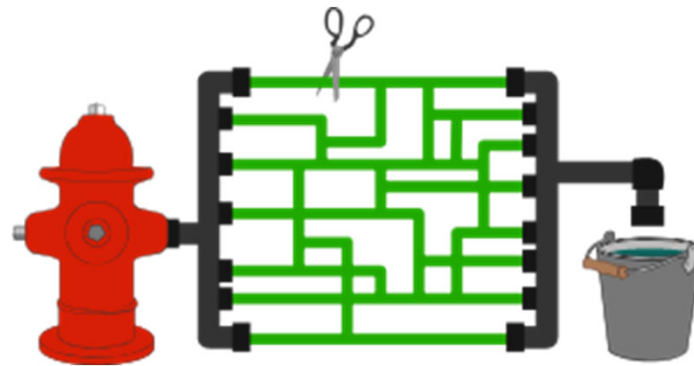
Edmonds-Karp Algorithm

Maximum Bipartite Matching

مبحث *Maximum Flow* از فصل ۲۶ کتاب *CLRS* تدریس می شود.

Maximum Flow

+a graph-theoretic definition of flow networks



A *flow network* $G = (V, E)$:

is a directed graph in which each edge $(u, v) \in E$ has a nonnegative *capacity* $c(u, v) \geq 0$.

If $(u, v) \notin E$, we assume that $c(u, v) = 0$.

In Flow networks

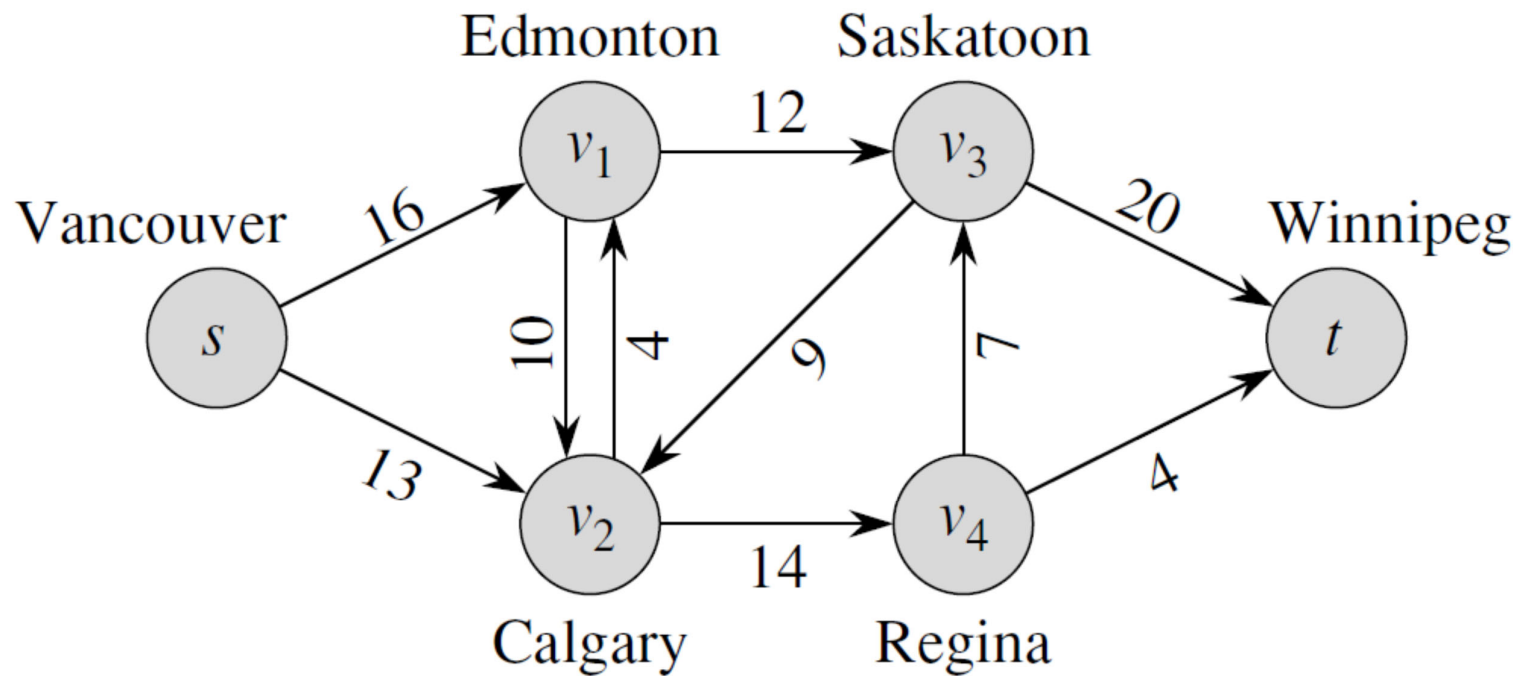
We distinguish : a **source** s and a **sink** t

we assume :

every vertex lies on some path from the source to the sink

The graph is therefore connected, and $|E| \geq |V| - 1$.

Example of network



flow

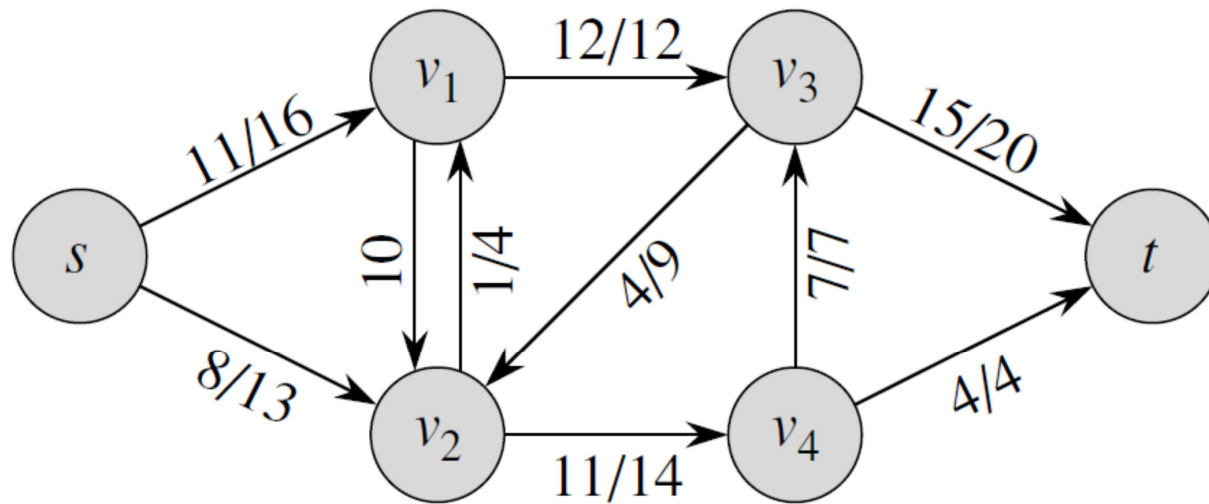
A flow in G is a **real-valued function** $f : V \times V \rightarrow R$ that satisfies the following three properties:

- **Capacity constraint:** For all $u, v \in V$, we require $f(u, v) \leq c(u, v)$.
- **Skew symmetry:** For all $u, v \in V$, we require $f(u, v) = -f(v, u)$.
- **Flow conservation:** For all $u \in V - \{s, t\}$, we require $\sum_{v \in V} f(u, v) = 0$
flow in equals **flow out** for vertex other than source and sink

The **value** of a flow f :

total flow out of the source ($|f| = \sum_{v \in V} f(s, v)$)

A sample flow



maximum-flow problem

In the ***maximum-flow problem***, we are given a flow network G with source s and sink t , and we wish to **find a flow of maximum value**.

Networks with multiple sources and sinks

- This problem is no harder than ordinary maximum flow
 - We can **reduce the problem** of determining a maximum flow in a network with multiple sources and multiple sinks to an ordinary maximum-flow problem

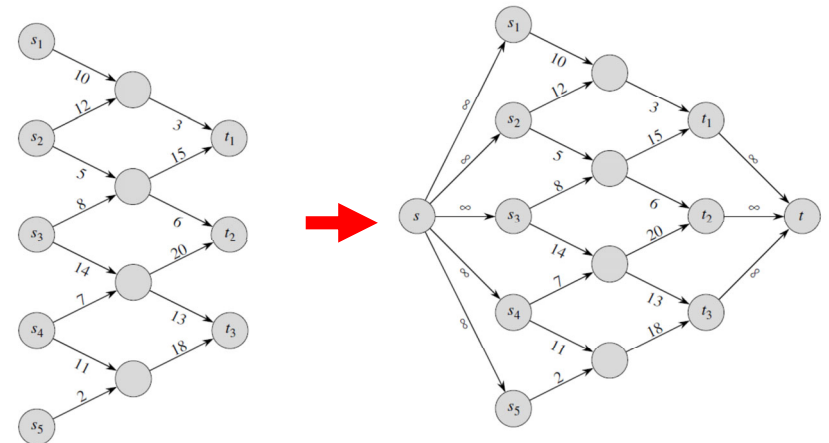
- **Add following:**

a **supersource** s

add a directed edge (s, s_i) with capacity $c(s, s_i) = \infty$ for each source s_i

a **supersink** t

and add a directed edge (t_i, t) with capacity $c(t_i, t) = \infty$ for each sink t_i



lemma

$$f(X, Y) = \sum_{x \in X} \sum_{y \in Y} f(x, y)$$

Let $G = (V, E)$ be a flow network, and let f be a flow in G . Then the following equalities hold:

1. For all $X \subseteq V$, we have $f(X, X) = 0$.
2. For all $X, Y \subseteq V$, we have $f(X, Y) = -f(Y, X)$.
3. For all $X, Y, Z \subseteq V$ with $X \cap Y = \emptyset$, we have the sums $f(X \cup Y, Z) = f(X, Z) + f(Y, Z)$ and $f(Z, X \cup Y) = f(Z, X) + f(Z, Y)$.

The value of flow

$$\begin{aligned}|f| &= f(s, V) \\ &= f(V, V) - f(V - s, V) \\ &= -f(V - s, V) \\ &= f(V, V - s) \\ &= f(V, t) + f(V, V - s - t) \\ &= f(V, t)\end{aligned}$$

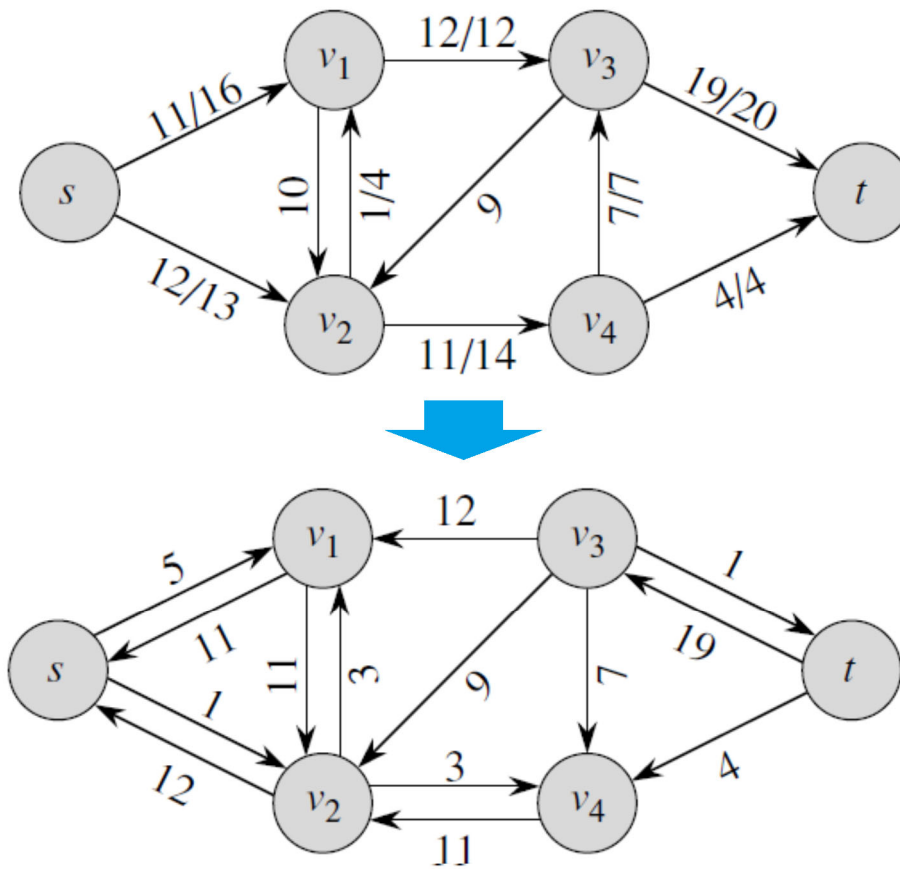
The Ford-Fulkerson method
+for solving the maximum-flow problem

Residual networks

- The amount of *additional* flow we can push from u to v before exceeding the capacity $c(u, v)$ is the ***residual capacity*** of (u, v) , given by

$$c_f(u, v) = c(u, v) - f(u, v).$$

given a flow network and a flow, the residual network consists of edges that can admit more flow



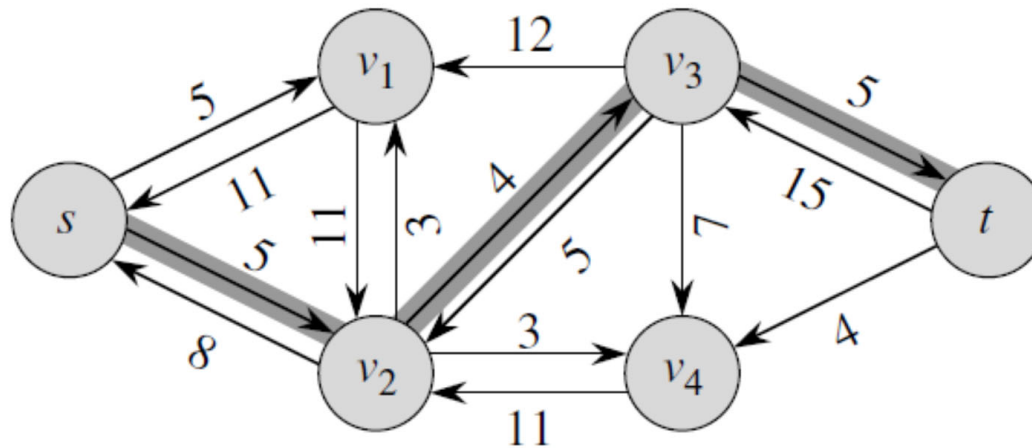
how a flow in a residual network relates to a flow in the original flow network

Let $G = (V, E)$ be a flow network with source s and sink t , and let f be a flow in G . Let G_f be the residual network of G induced by f , and let f' be a flow in G_f . Then the flow sum $f + f'$ defined by equation (26.4) is a flow in G with value $|f + f'| = |f| + |f'|$.

Augmenting paths

Augmenting path is

a path from the source s to the sink t along which we can send more flow, and then augmenting the flow along this path



❖ A flow is maximum if and only if its residual network contains no augmenting path.

Residual capacity of an augmenting path

: The maximum amount by which we can increase the flow on each edge in an augmenting path p

$$c_f(p) = \min \{c_f(u, v) : (u, v) \text{ is on } p\}$$

residual capacity of an augmenting path can add to the value of flow

Let $G = (V, E)$ be a flow network, let f be a flow in G , and let p be an augmenting path in G_f . Define a function $f_p : V \times V \rightarrow \mathbf{R}$ by

$$f_p(u, v) = \begin{cases} c_f(p) & \text{if } (u, v) \text{ is on } p, \\ -c_f(p) & \text{if } (v, u) \text{ is on } p, \\ 0 & \text{otherwise.} \end{cases} \quad (26.6)$$

Then, f_p is a flow in G_f with value $|f_p| = c_f(p) > 0$.

Corollary

Let $G = (V, E)$ be a flow network, let f be a flow in G , and let p be an augmenting path in G_f . Let f_p be defined as in equation (26.6). Define a function $f' : V \times V \rightarrow \mathbf{R}$ by $f' = f + f_p$. Then f' is a flow in G with value $|f'| = |f| + |f_p| > |f|$.

General method

How increase the value of flow

FORD-FULKERSON-METHOD(G, s, t)

1 initialize flow f to 0

2 **while** there exists an augmenting path p

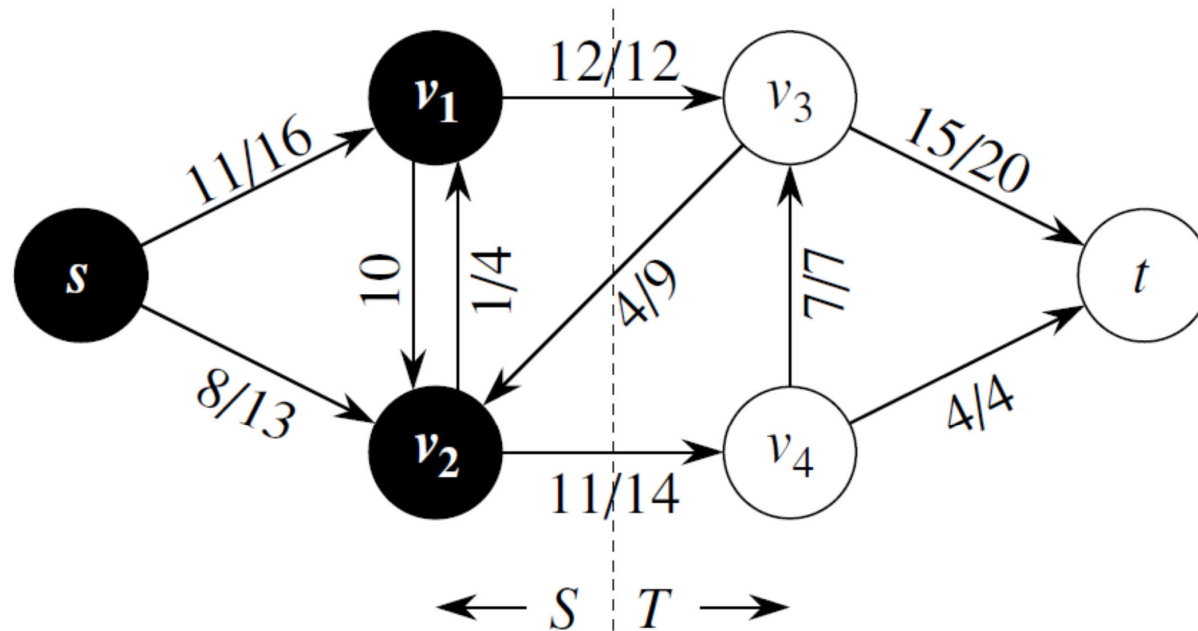
3 **do** augment flow f along p

4 **return** f

Definition of cut

- A **cut** (S, T) of flow network $G = (V, E)$ is a partition of V into S and $T = V - S$ such that $s \in S$ and $t \in T$.
- If f is a flow, then the **net flow** across the cut (S, T) is defined to be $f(S, T)$.
- The **capacity** of the cut (S, T) is $c(S, T)$.
- A **minimum cut** of a network is a cut whose capacity is minimum over all cuts of the network.

A cut with
the net flow $f(S, T) = 19$, and
the capacity $c(S, T) = 26$.



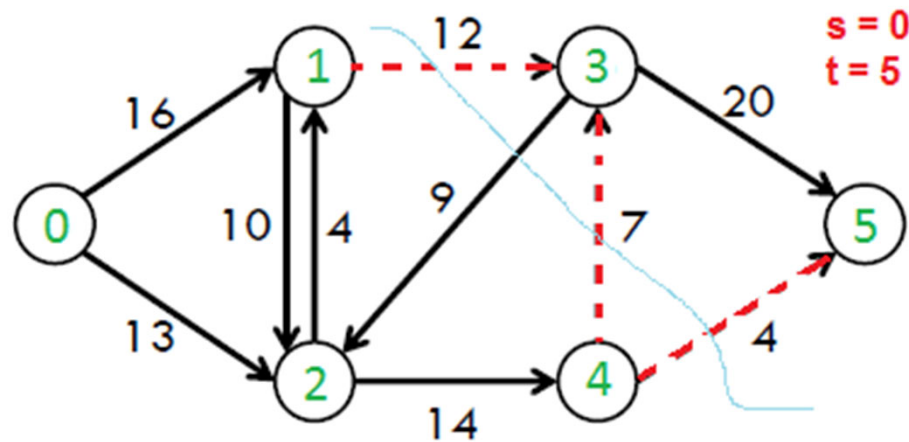
- The value of any flow f in a flow network G is bounded from above by the capacity of any cut of G .

(can proved by definition of flow and cut)

Max-flow min-cut theorem

If f is a flow in a flow network $G = (V, E)$ with source s and sink t , then the following conditions are equivalent:

1. f is a maximum flow in G .
2. The residual network G_f contains no augmenting paths.
3. $|f| = c(S, T)$ for some cut (S, T) of G .



Max-flow min-cut theorem

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Corollary

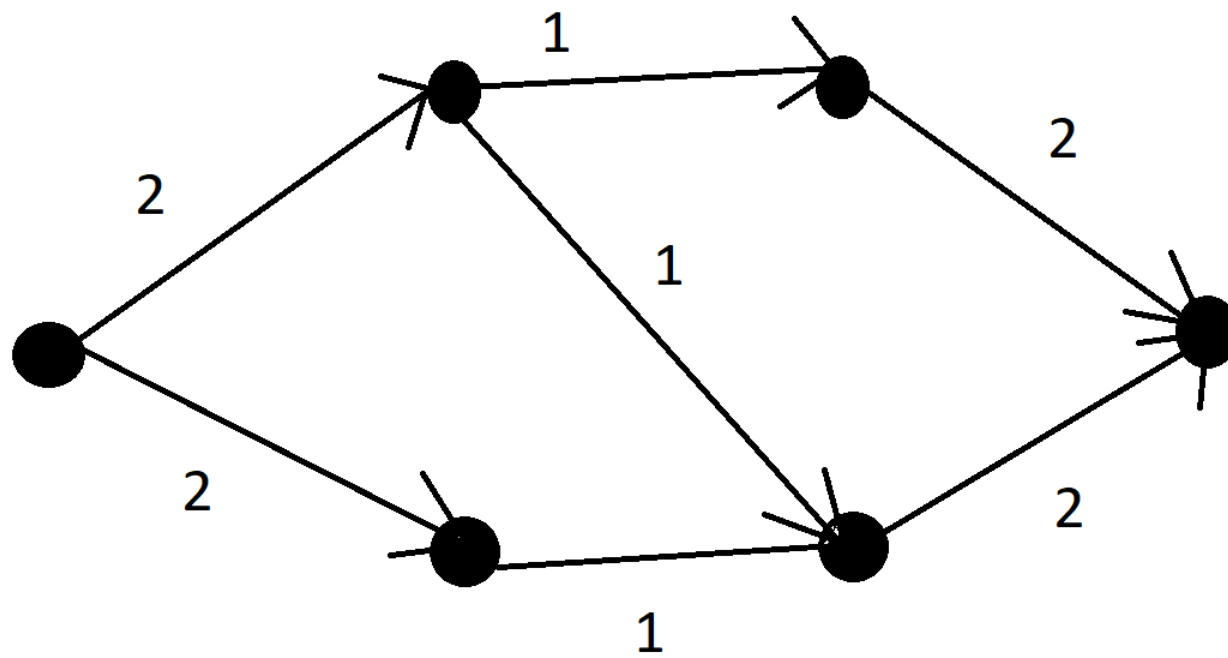
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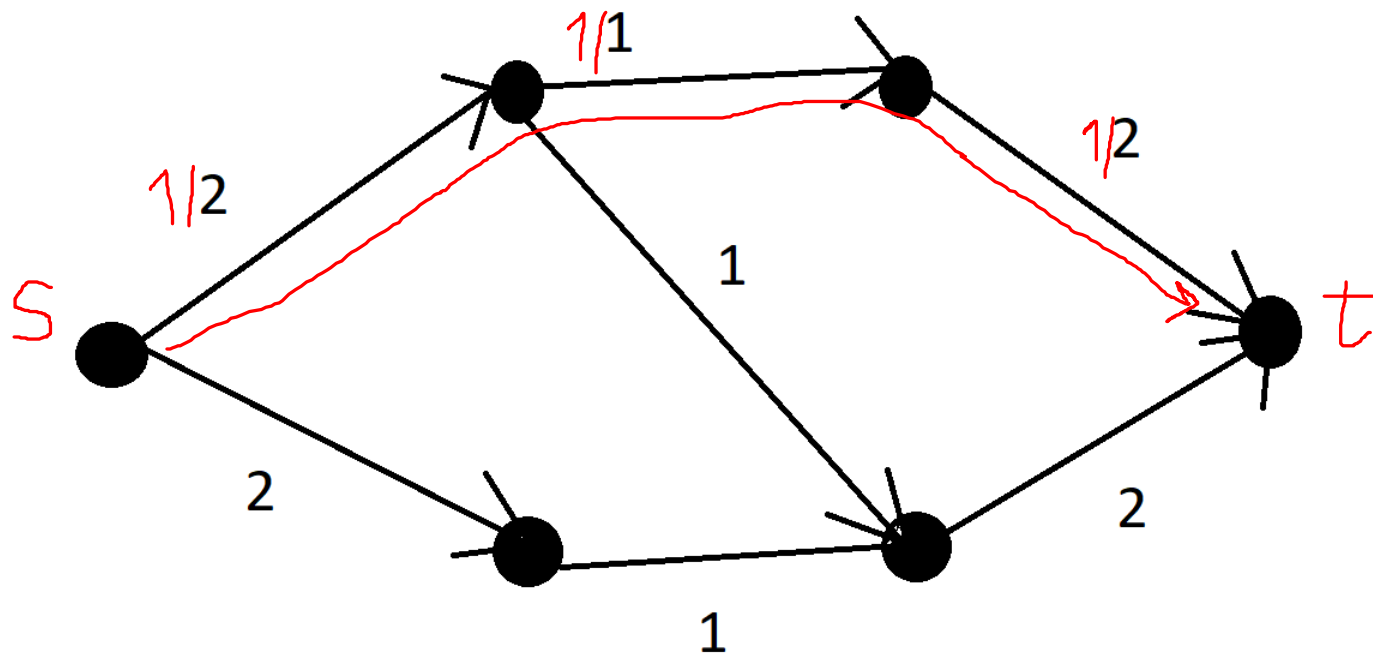
basic Ford-Fulkerson algorithm

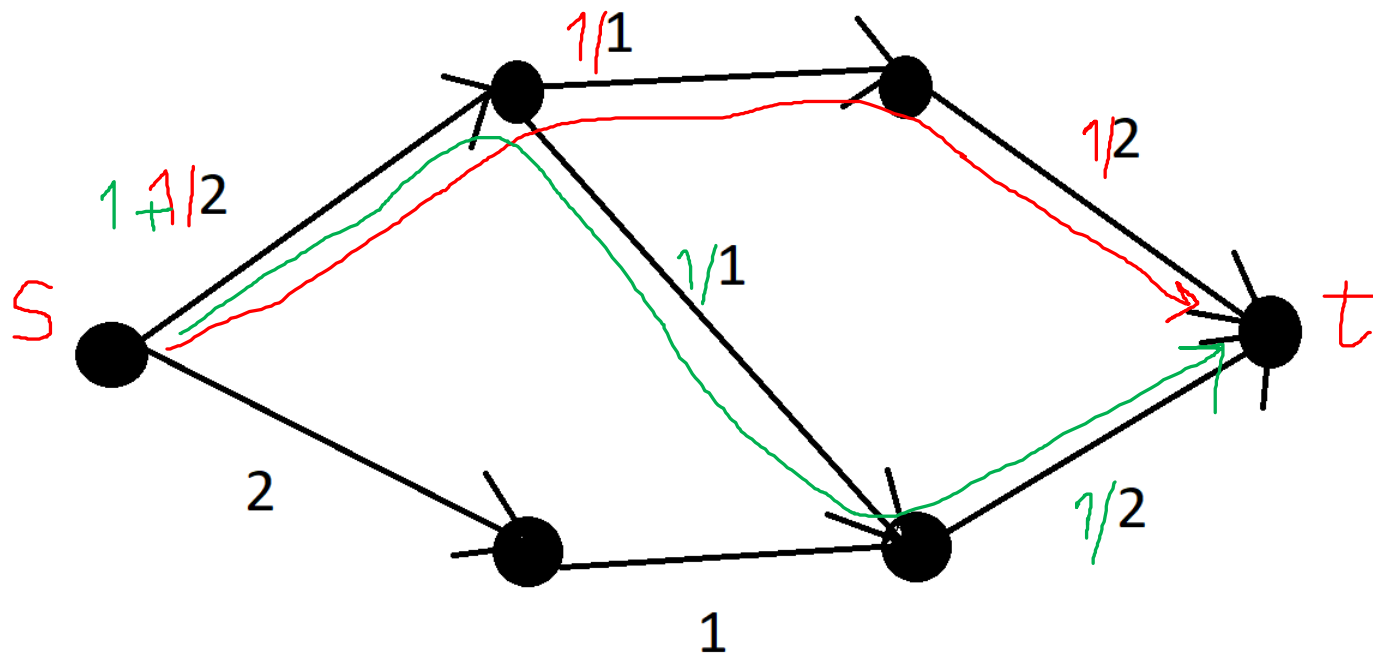
expands on the FORD-FULKERSON METHOD

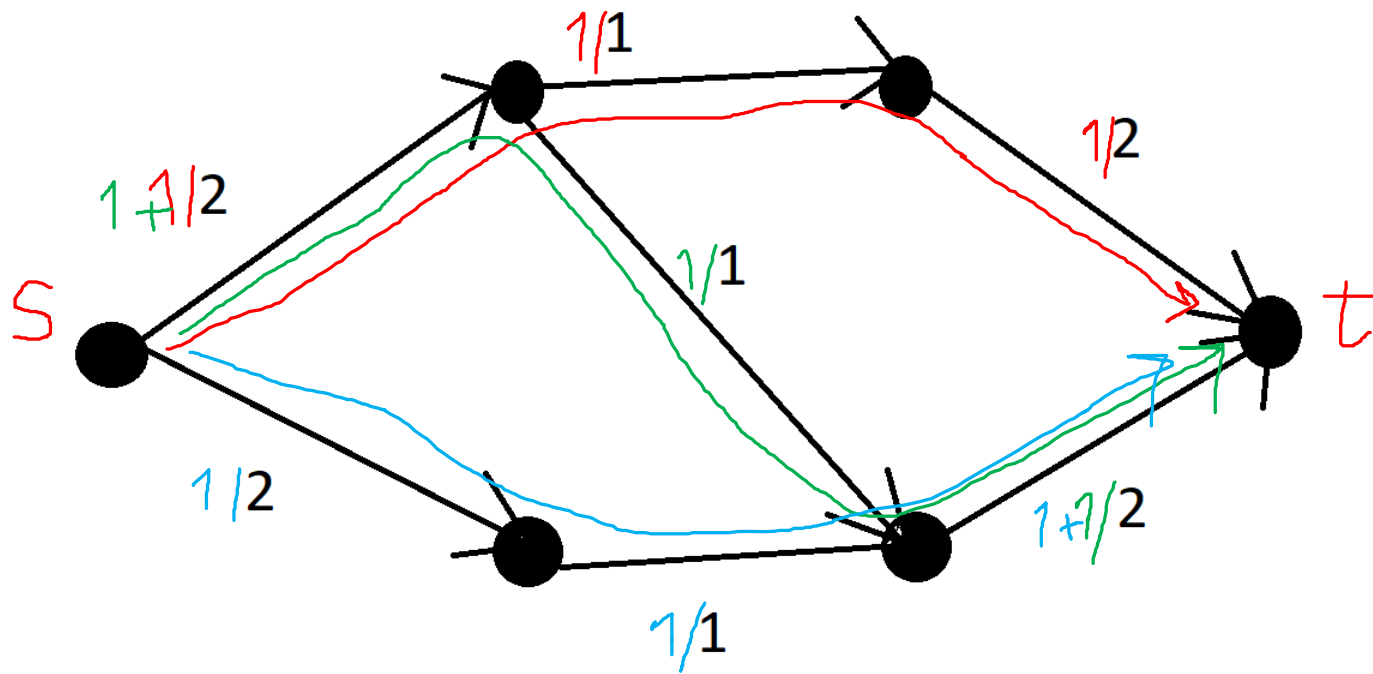
FORD-FULKERSON(G, s, t)

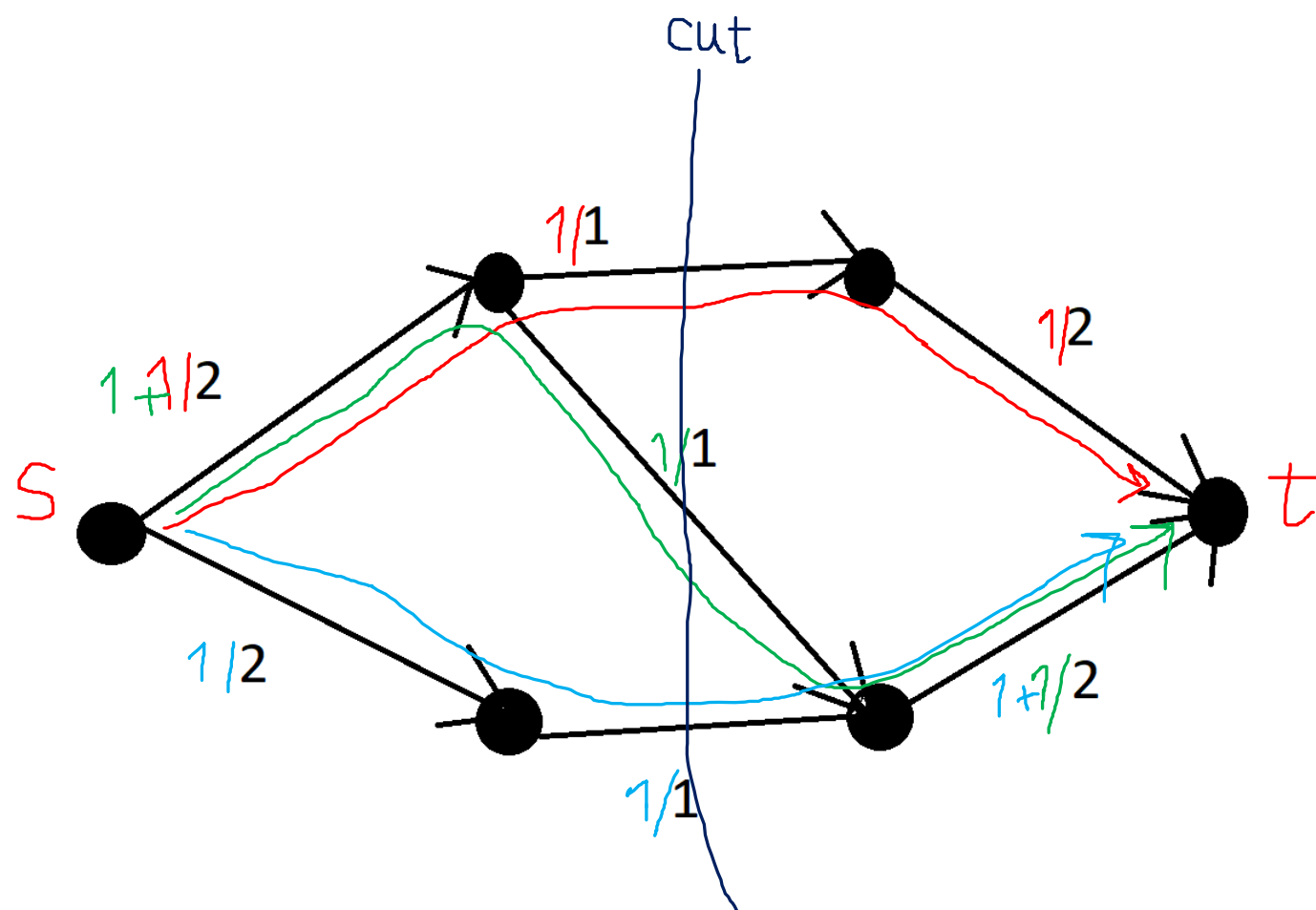
```
1  for each edge  $(u, v) \in E[G]$ 
2      do  $f[u, v] \leftarrow 0$ 
3       $f[v, u] \leftarrow 0$ 
4  while there exists a path  $p$  from  $s$  to  $t$  in the residual network  $G_f$ 
5      do  $c_f(p) \leftarrow \min \{c_f(u, v) : (u, v) \text{ is in } p\}$ 
6      for each edge  $(u, v)$  in  $p$ 
7          do  $f[u, v] \leftarrow f[u, v] + c_f(p)$ 
8           $f[v, u] \leftarrow -f[u, v]$ 
```

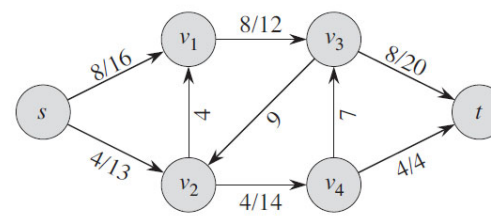
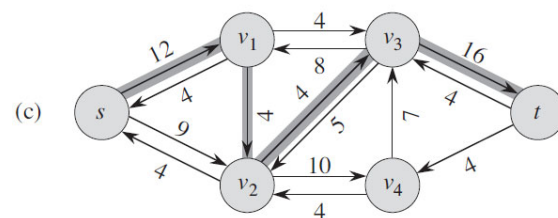
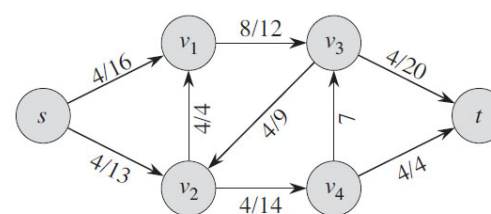
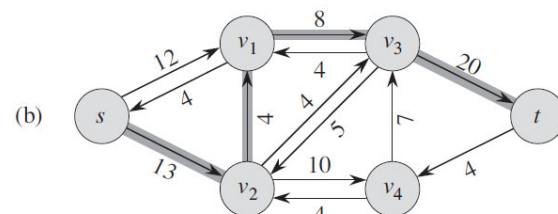
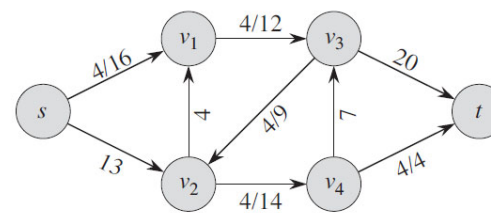
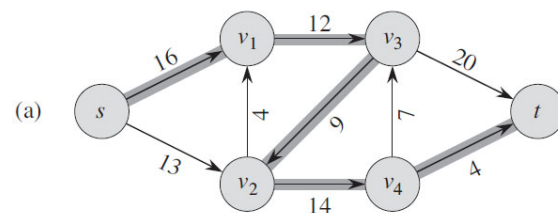


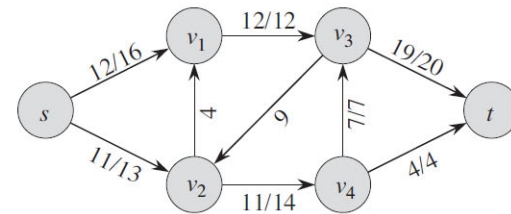
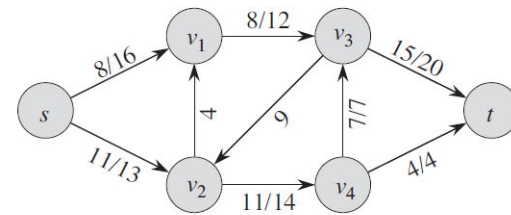
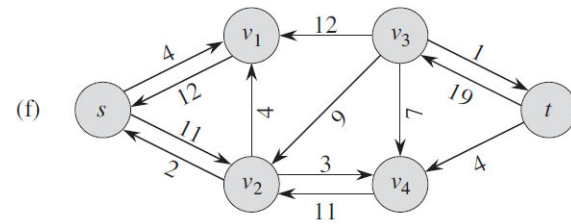
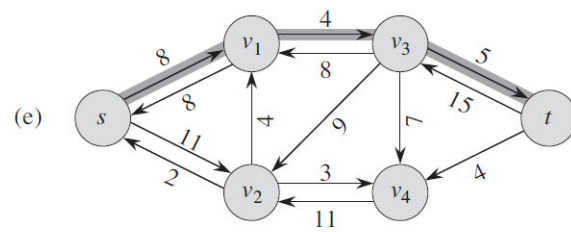
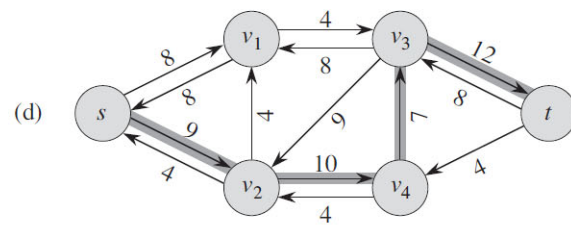












Time complexity

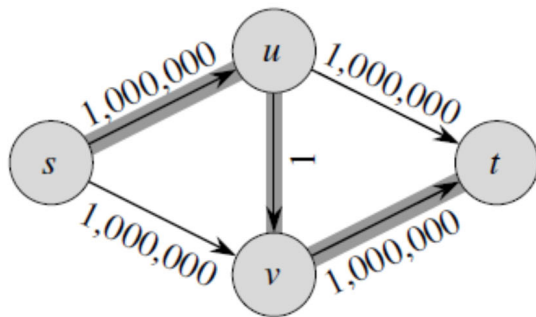
- The running time of FORD-FULKERSON depends on how the augmenting path p in line 4 is determined.

a straightforward implementation runs in time $O(E |f^*|)$

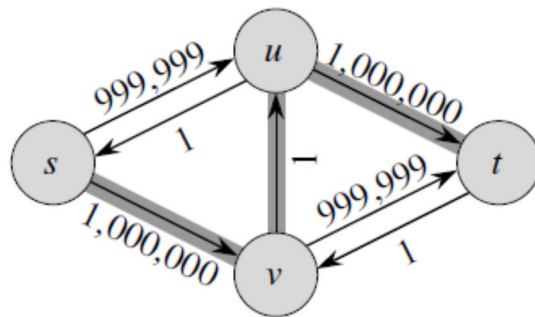
- Prove Hint:
the flow value increases by at least one unit in each iteration

$O(E |f^*|)$ can be bad!

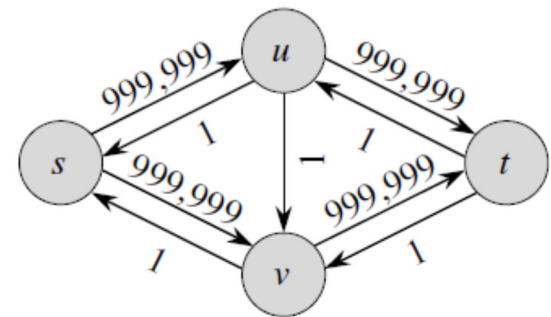
- continue, choosing the augmenting path $s \rightarrow u \rightarrow v \rightarrow t$ in the odd-numbered iterations and the augmenting path $s \rightarrow v \rightarrow u \rightarrow t$ in the even-numbered iterations.



(a)



(b)



(c)

The Edmonds-Karp algorithm

- The bound on FORD-FULKERSON can be improved if we implement the computation of the augmenting path p in line 4 **with a breadth-first search** (each edge has unit distance (weight))

the augmenting path is a *shortest* path from s to t in the residual network

the running time of the Edmonds-Karp algorithm is $O(V E^2)$

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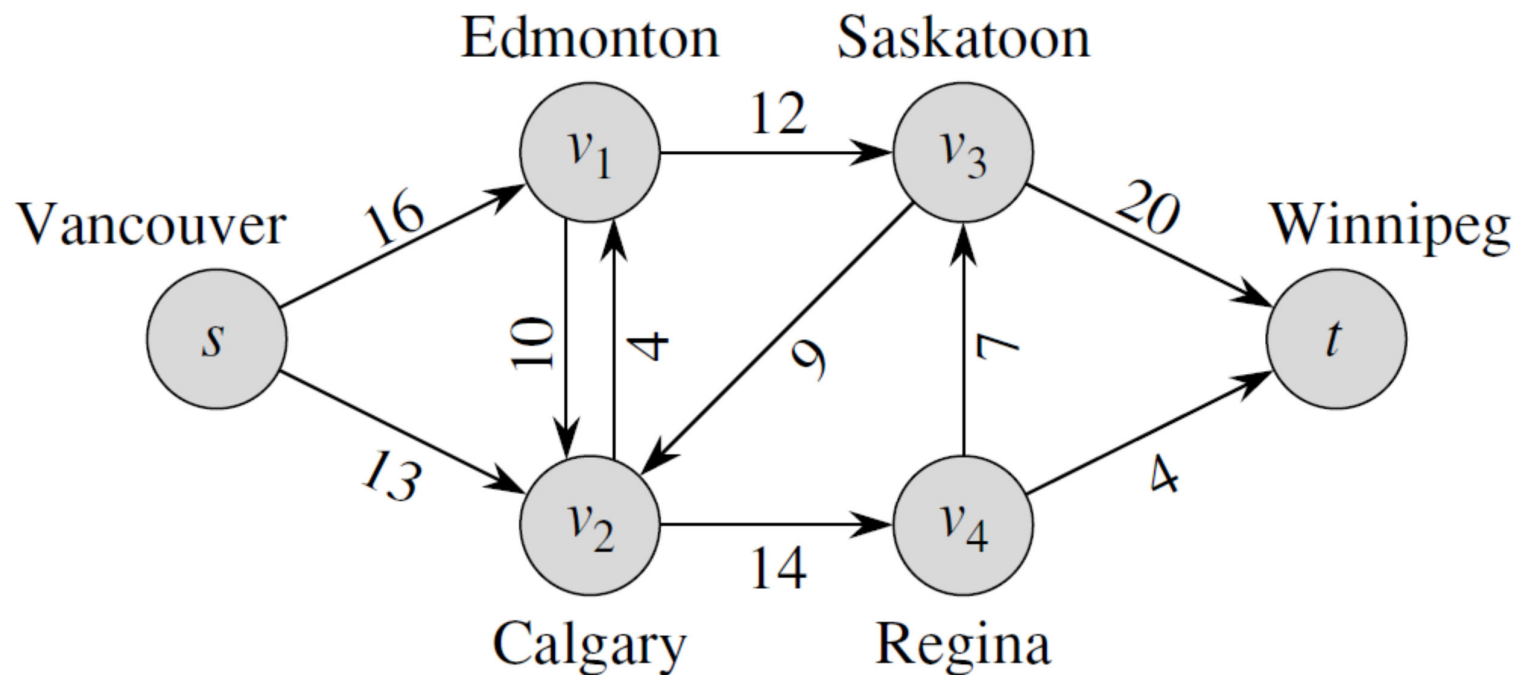
Lemma 26.8

If the Edmonds-Karp algorithm is run on a flow network $G = (V, E)$ with source s and sink t , then for all vertices $v \in V - \{s, t\}$, the shortest-path distance $\delta_f(s, v)$ in the residual network G_f increases monotonically with each flow augmentation.

Theorem 26.9

If the Edmonds-Karp algorithm is run on a flow network $G = (V, E)$ with source s and sink t , then the total number of flow augmentations performed by the algorithm is $O(V E)$.

Show the execution of the Edmonds-Karp algorithm on the flow network



Sample Problem

Suppose you are given a flow network G with *integer* edge capacities and an *integer* maximum flow f^* in G . Describe algorithms for the following operations:

- (a) INCREMENT(e): Increase the capacity of edge e by 1 and update the maximum flow.
- (b) DECREMENT(e): Decrease the capacity of edge e by 1 and update the maximum flow.

Both algorithms should modify f^* so that it is still a maximum flow, more quickly than recomputing a maximum flow from scratch.

Maximum bipartite matching
+solve using flow network

The maximum matching problem

Given an undirected graph $G = (V, E)$,

Matching is a subset of edges $M \subseteq E$ such that for all vertices $v \in V$, at most one edge of M is incident on v .

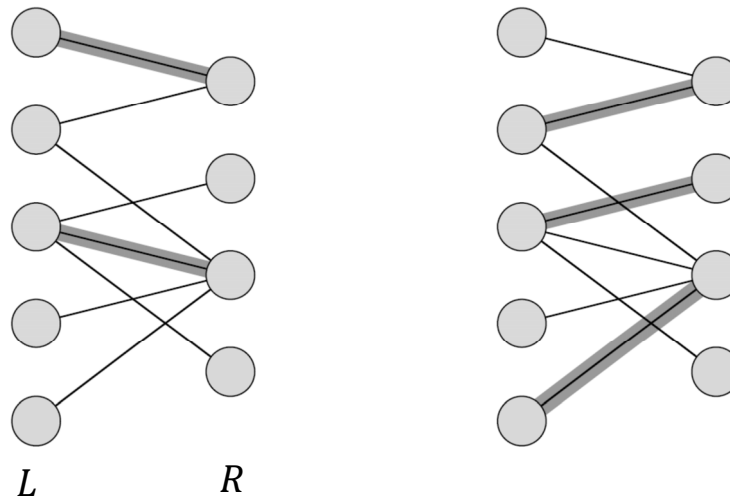
Maximum matching is a matching of maximum cardinality



a vertex $v \in V$ is **matched** by matching M if some edge in M is incident on v ; otherwise, v is **unmatched**.

The maximum-bipartite-matching problem

- **Bipartite graphs:** the vertex set can be partitioned into $V = L \cup R$, where L and R are **disjoint** and **all edges** in E go **between L and R** .
(further assume that every vertex in V has at least one incident edge.)



Finding a maximum bipartite matching

- We can **use the Ford-Fulkerson method** to find a maximum matching in an **undirected bipartite graph** $G = (V, E)$ in time **polynomial in $|V|$ and $|E|$** .

- First:

define the ***corresponding flow network*** $G' = (V', E')$ for the bipartite graph G

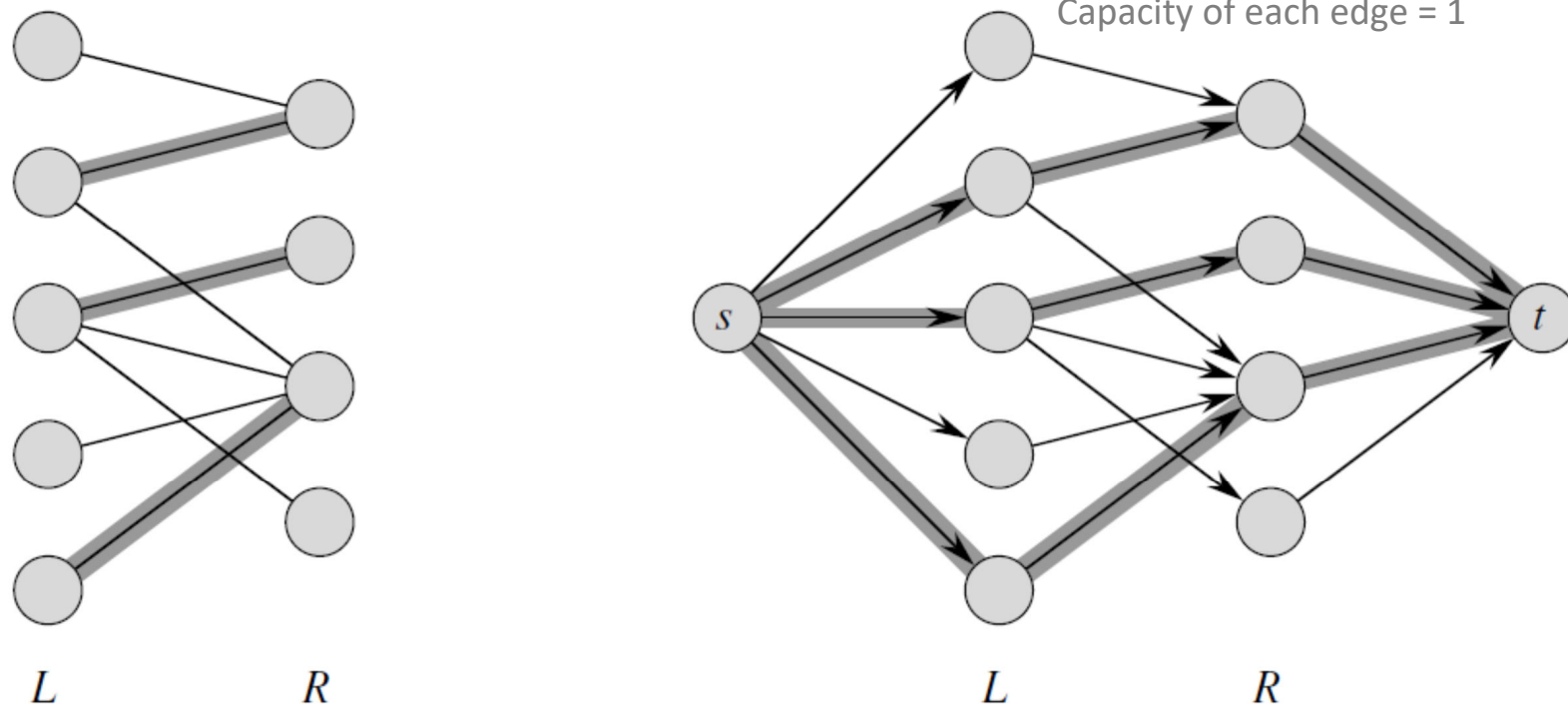
corresponding flow network

- Add source s and sink t as new vertices
 - let $V' = V \cup \{s, t\}$.
- The directed edges of G' are :
 - the edges of E , directed from L to R ($V = L \cup R$), along with V new edges:

$$\begin{aligned} E' = & \{(s, u) : u \in L\} \\ & \cup \{(u, v) : u \in L, v \in R, \text{ and } (u, v) \in E\} \\ & \cup \{(v, t) : v \in R\} . \end{aligned}$$

- assign unit capacity to each edge in E' .

The flow network corresponding to a bipartite graph



The order of E' ?

$$|E| \leq |E'|$$

$$|E'| = |E| + |V| \leq 3|E|, (|E| \geq |V|/2)$$

$$\text{so } |E'| = \Theta(E)$$

a matching in G corresponds directly to
a flow in G 's corresponding flow network

Lemma 26.10

Let $G = (V, E)$ be a bipartite graph with vertex partition $V = L \cup R$, and let $G' = (V', E')$ be its corresponding flow network. If M is a matching in G , then there is an integer-valued flow f in G' with value $|f| = |M|$. Conversely, if f is an integer-valued flow in G' , then there is a matching M in G with cardinality $|M| = |f|$.

maximum matching in a bipartite graph *and* the value of a maximum flow

Theorem 26.11 (Integrality theorem)

If the capacity function c takes on only integral values, then the maximum flow f produced by the Ford-Fulkerson method has the property that $|f|$ is integer-valued. Moreover, for all vertices u and v , the value of $f(u, v)$ is an integer.

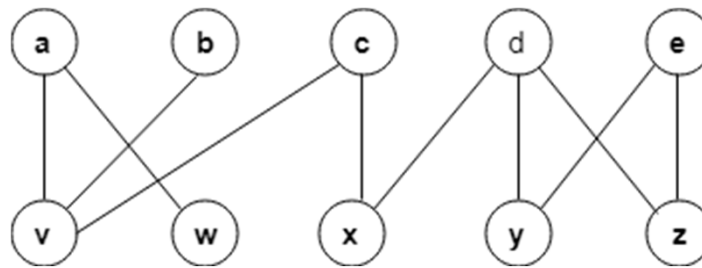
Corollary 26.12

The cardinality of a maximum matching M in a bipartite graph G is the value of a maximum flow f in its corresponding flow network G' .

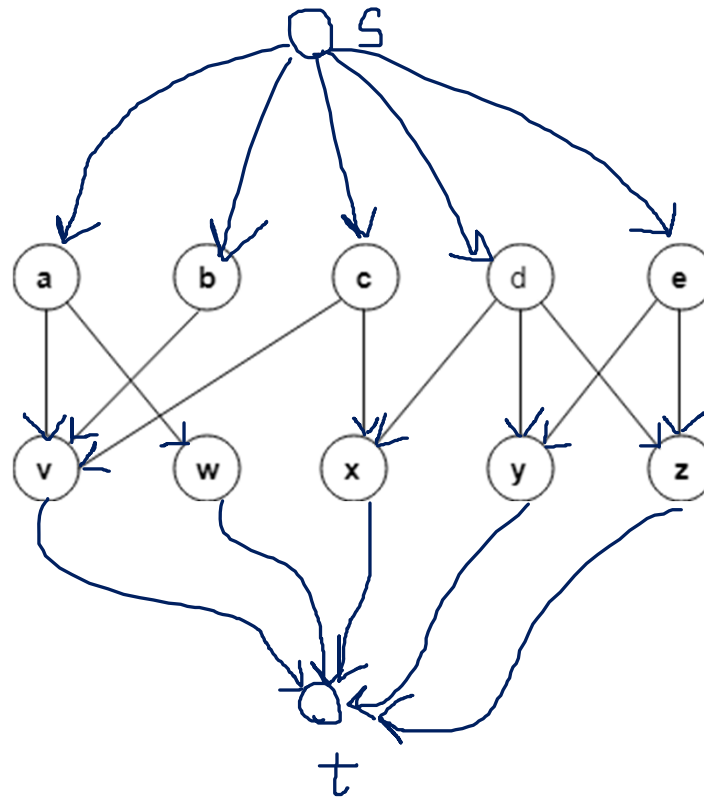
Whole picture

- Given a bipartite undirected graph G ,
 - Create the flow network G'
 - Run the Ford-Fulkerson method
 - Obtaining a maximum matching M from the integer-valued maximum flow
- Ford-Fulkerson time complexity = $O(E |f^*|)$
- The value of the maximum flow in G' is $O(V)$.
 - Since any matching in a bipartite graph has cardinality at most $\min(L, R) = O(V)$,
Time complexity of finding a maximum matching in a bipartite graph ($|E'| = \Theta(E)$)
is $O(V E)$.

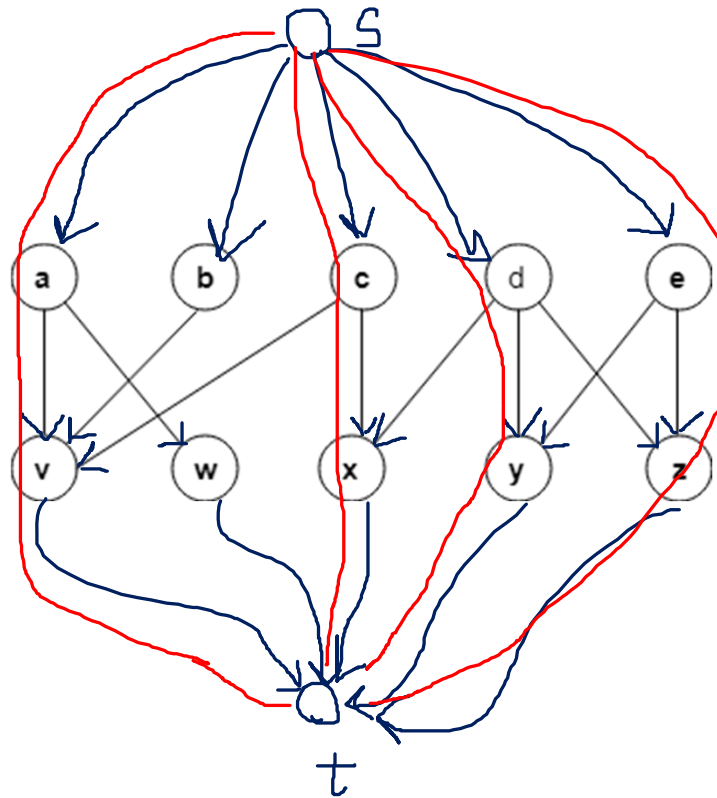
Compute the maximum bipartite matching



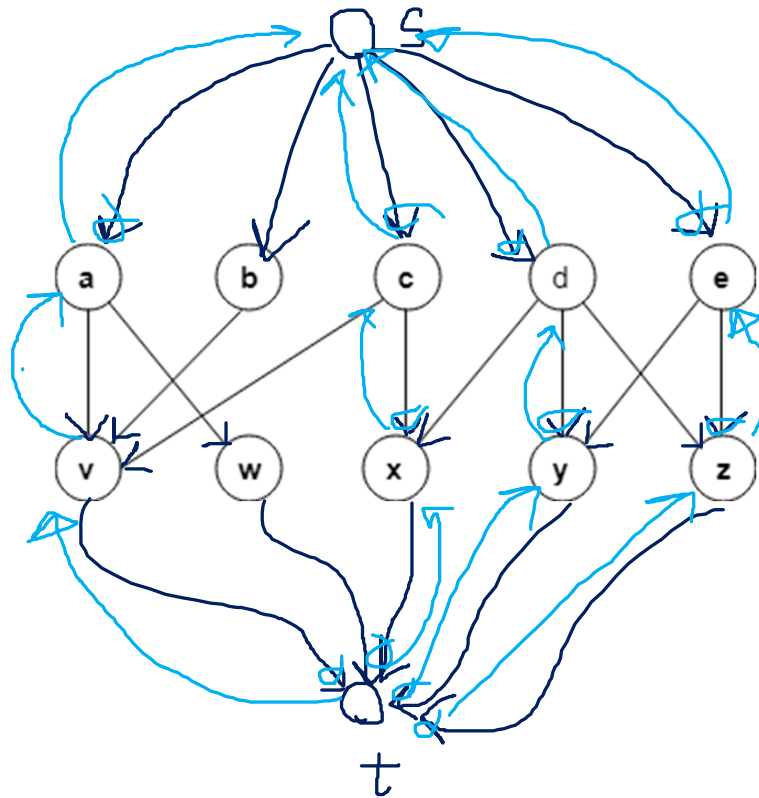
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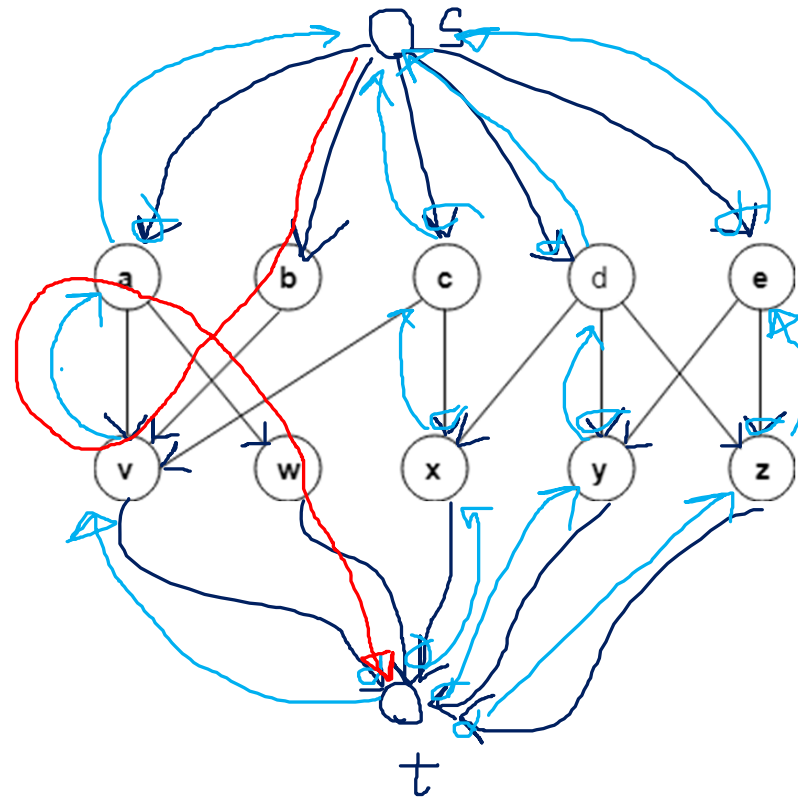
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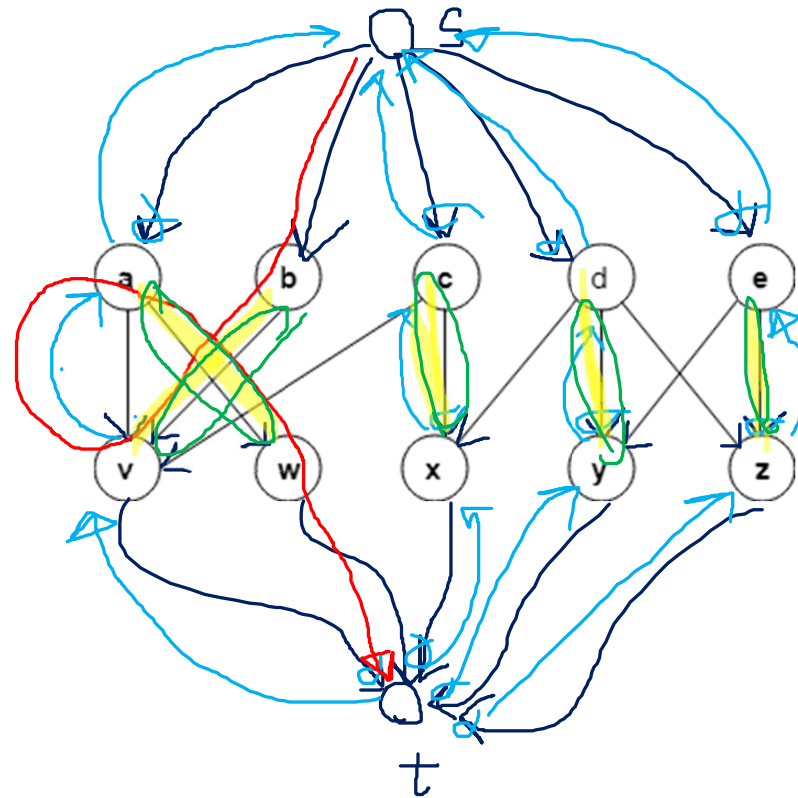
Compute the maximum bipartite matching



Compute the maximum bipartite matching



Compute the maximum bipartite matching



Some sample problems

1- If all edges in a graph have distinct capacities, is there a unique maximum flow?

2- Let $G = (V, E)$ be a bipartite graph with vertex partition $V = L \cup R$, and let G' be its corresponding flow network. Give a good upper bound on the length of any augmenting path found in G' during the execution of FORD-FULKERSON.

3- A **perfect matching** is a matching in which every vertex is matched.

Let $G = (V, E)$ be an undirected bipartite graph with vertex partition $V = L \cup R$, where $|L| = |R|$. For any $X \subseteq V$, define the **neighborhood** of X as $N(X) = \{y \in V : (x, y) \in E \text{ for some } x \in X\}$, that is, the set of vertices adjacent to some member of X .

Prove **Hall's theorem**: there exists a perfect matching in G if and only if $|A| \leq |N(A)|$ for every subset $A \subseteq L$.

Hint : use cut